



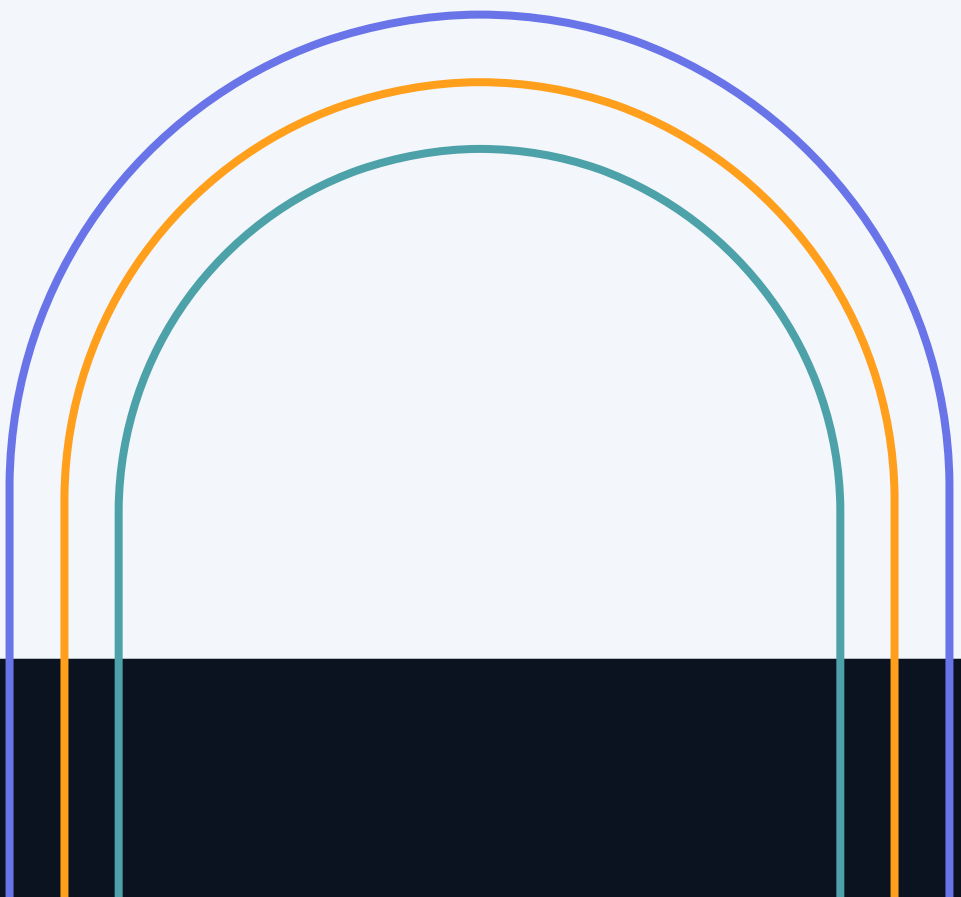


LLMs:

How LLMs answer

prompts

— Nicholas Ramli



Intro: LLMs



Large Language Models

- Predict the next word of sentence to give a response
- Able to make predictions by being trained on large amounts of text

How LLMs Work

- Words from prompt turned into tokens:
 - Response is generated from next, most probable token

Context

The passage given by the user will be completed with the correct following lines from the novel.

User-Input

There now is your insular city of the Manhattoes, belted round by wharves as Indian isles by coral reefs--commerce surrounds it with her surf.

Prompt

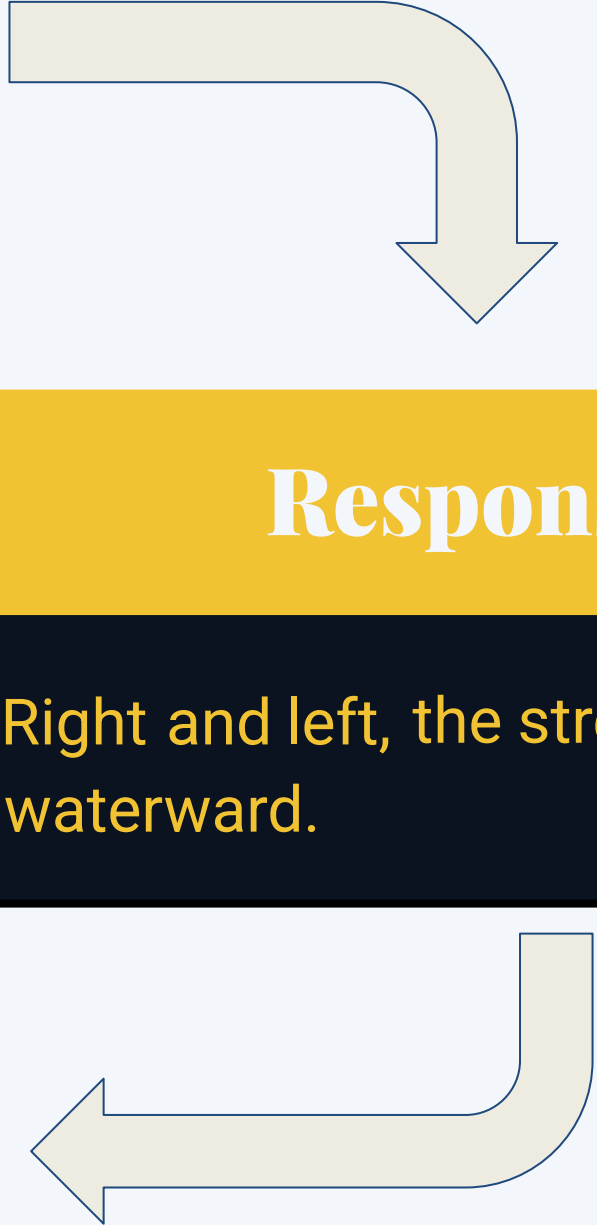
The passage given by the user will be completed with the correct following lines from the novel.

There now is your insular city of the Manhattoes, belted round by wharves as Indian isles by coral reefs--commerce surrounds it with her surf.

Right and left, the streets take you waterward.

Response

Right and left, the streets take you waterward.



```
graph LR; Context[Context] --> Prompt[Prompt]; User-Input[User-Input] --> Prompt; Prompt --> Response[Response]
```

Training



Pre-training

- Starts as random guesses
- Adjusts parameters to generate actual text

Fine-Tuning

- Trained on specialized datasets
- Shows LLM exact type of input and output it should produce

RLHF

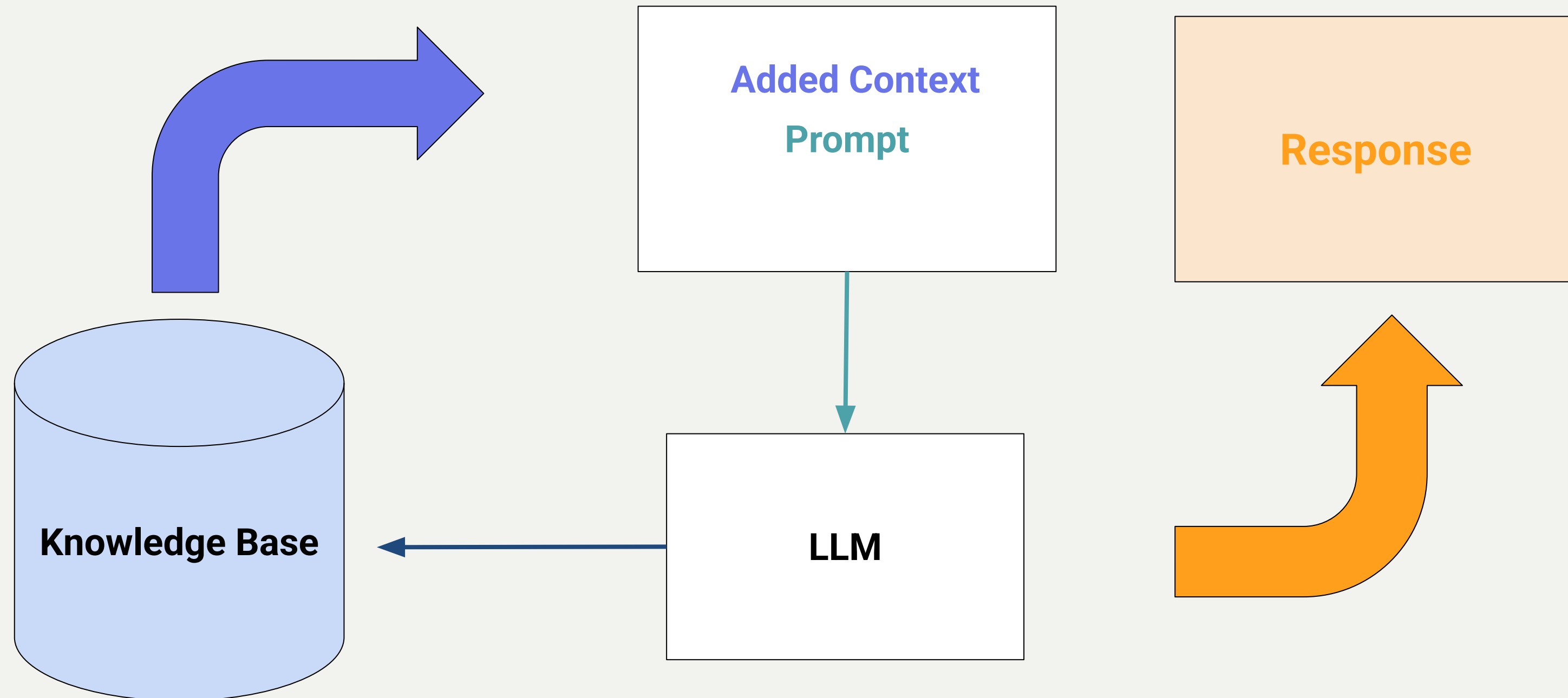
- Reinforcement learning from human feedback
- Output judged by people as helpful or unhelpful



How LLMs handle information outside of training data

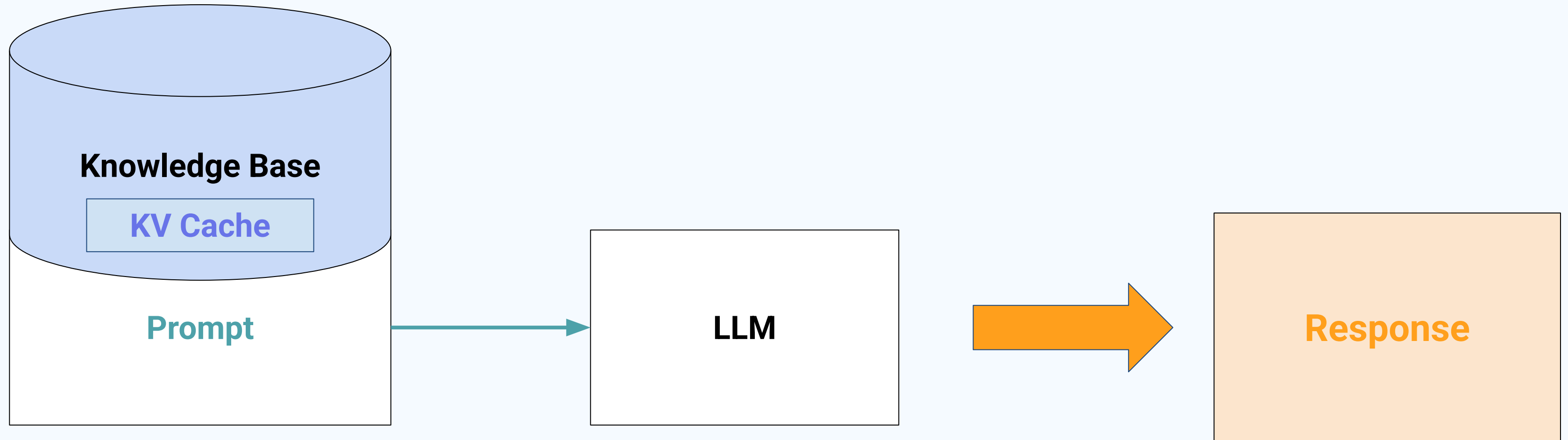
RAG

Retrieval Augmented Generation



CAG

Cache Augmented Generation



Benefits of RAG

1

- **Can have large data sets**

2

- **Update new data easily**

3

- **Less unrelated data**

Benefits of CAG

1

- **Faster, no need to retrieve data**

2

- **Great for static knowledge bases**

3

- **Will have relevant data**